



MS/MA IN ECONOMICS

DYNAMIC STOCHASTIC MODELS

Lecture 10

Small Open Economy Models

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The Homework

Consider a small open economy without money in which there are shocks to the foreign interest rate r^* . The foreign interest rate is now written as

$$r_t^* = \varphi r^* + (1 - \varphi) r_{t-1}^* + \varepsilon_t^r, \quad \varphi < 1, \quad (1)$$

so that the base foreign interest rate is no longer a constant but follows a stochastic process around its long-run value r^* . The tasks are:

1. Write out the model. The stationary states are the same as in class.
2. Write out the log-linear version of the model.
3. Write out the matrices for solving the model.
4. Use `llinsolve` to find the P, Q, R, S matrices and the impulse response functions for a shock to the foreign interest rate.

This exercise sits in Chapter 13 of McCandless (2008), *The ABCs of the RBCs*, Section 13.3 (“Closing the Open Economy”). The model we work with is the closed open economy without money, which includes capital adjustment costs and an endogenous foreign interest rate. The only departure from the model solved in class is that r^* becomes stochastic, which introduces a second shock into the system in addition to the technology shock λ_t .

1 The Model

1.1 The Household

The representative household chooses sequences $\{c_t, h_t, k_{t+1}, b_t\}_{t=0}^{\infty}$ to maximize expected discounted utility

$$E_t \sum_{t=0}^{\infty} \beta^t [\ln c_t + B h_t] \quad (2)$$

subject to the flow budget constraint, which includes capital adjustment costs and the endogenous foreign interest rate,

$$b_t + k_{t+1} + \frac{\kappa}{2} (k_{t+1} - k_t)^2 + c_t = w_t h_t + r_t k_t + (1 - \delta) k_t + \left(1 + r_{t-1}^f\right) b_{t-1}, \quad (3)$$

where $r_t^f = r_t^* - aB_t$ is the foreign interest rate faced by the country. The Lagrangian is

$$\mathcal{L} = E_t \sum_{t=0}^{\infty} \beta^t \left\{ \ln c_t + Bh_t + \mu_t \left[w_t h_t + r_t k_t + (1 - \delta)k_t + \left(1 + r_{t-1}^f\right) b_{t-1} - b_t - k_{t+1} - \frac{\kappa}{2} (k_{t+1} - k_t)^2 - c_t \right] \right\} \quad (4)$$

We do not replace r^f in the Lagrangian. Since the interest rate depends on the country's total borrowing or savings, rather than on the borrowing or savings of an individual family, the first-order conditions of the family are not affected by this relationship. Only after obtaining the first-order conditions do we substitute the expression for r_t^f . The derivatives of the Lagrangian are

$$\frac{\partial \mathcal{L}}{\partial c_t} = \frac{1}{c_t} - \mu_t = 0 \quad \Rightarrow \quad \mu_t = \frac{1}{c_t}, \quad (5)$$

$$\frac{\partial \mathcal{L}}{\partial h_t} = B + \mu_t w_t = 0 \quad \Rightarrow \quad B = -\mu_t w_t, \quad (6)$$

$$\frac{\partial \mathcal{L}}{\partial b_t} = -\mu_t + \beta E_t \mu_{t+1} (1 + r_t^f) = 0 \quad \Rightarrow \quad \mu_t = \beta E_t \mu_{t+1} (1 + r_t^f) \quad (7)$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = -\mu_t [1 + \kappa (k_{t+1} - k_t)] + \beta E_t \mu_{t+1} [r_{t+1} + (1 - \delta) + \kappa (k_{t+2} - k_{t+1})] = 0 \quad (8)$$

$$\Rightarrow \quad \frac{1}{\beta} [1 + \kappa (k_{t+1} - k_t)] = E_t \frac{\mu_{t+1}}{\mu_t} [r_{t+1} + (1 - \delta) + \kappa (k_{t+2} - k_{t+1})]. \quad (9)$$

Substituting $\mu_t = 1/c_t$ into (5), (6) and (9), the family first-order conditions become

$$B = -\frac{w_t}{c_t}, \quad (10)$$

$$\frac{1}{c_t} = \beta E_t \frac{1}{c_{t+1}} (1 + r_t^f), \quad (11)$$

$$\frac{1}{\beta} [1 + \kappa (k_{t+1} - k_t)] = E_t \frac{c_t}{c_{t+1}} [r_{t+1} + (1 - \delta) + \kappa (k_{t+2} - k_{t+1})], \quad (12)$$

together with the flow budget constraint (3).

1.2 The Firm

Domestic firms are perfectly competitive and operate a Cobb–Douglas technology,

$$Y_t = \lambda_t K_t^\theta H_t^{1-\theta}, \quad (13)$$

where λ_t is total factor productivity and $\theta \in (0, 1)$ is the capital share. The technology shock follows the autoregressive process

$$\lambda_{t+1} = \gamma \lambda_t + \varepsilon_{t+1}, \quad 0 < \gamma < 1, \quad E_t \varepsilon_{t+1} = 1 - \gamma. \quad (14)$$

Taking factor prices as given, profit maximization delivers the factor demands, which equate each factor price to its marginal product,

$$r_t = \theta \lambda_t K_t^{\theta-1} H_t^{1-\theta}, \quad (15)$$

$$w_t = (1 - \theta) \lambda_t K_t^\theta H_t^{-\theta}. \quad (16)$$

1.3 Closing the Open Economy

To close the model we introduce the concept of country risk through

$$r_t^f = r_t^* - a B_t, \quad a > 0, \quad (17)$$

where a is a positive constant, so that as the country accumulates foreign debt (B_t more negative) the interest rate it must pay rises. If the international interest rate were constant, we would have to impose the no-Ponzi-game condition

$$\lim_{t \rightarrow \infty} \beta^t b_t = 0. \quad (18)$$

However, under the country-risk rule (17) the interest rate rises as net foreign debt increases, which makes borrowing more expensive and prevents the household from accumulating debt indefinitely. Unlike the version solved in class, the base rate r^* is no longer constant: it follows the autoregressive process

$$r_t^* = \varphi r^* + (1 - \varphi) r_{t-1}^* + \varepsilon_t^r, \quad \varphi < 1, \quad (19)$$

where r^* is the long-run value of the base rate and ε_t^r is a mean-zero innovation. In the stationary state $r_t^* = r^*$, so this process introduces a second source of uncertainty without altering the stationary state.

1.4 First Order and Equilibrium Conditions

Since all families in the unit mass are identical, aggregate variables are given by

$$C_t = \int_0^1 c_t(i) di = c_t, \quad K_t = \int_0^1 k_t(i) di = k_t, \quad H_t = \int_0^1 h_t(i) di = h_t, \quad B_t = \int_0^1 b_t(i) di = b_t. \quad (20)$$

Replacing the aggregation conditions (20) into the family equilibrium conditions, the conditions for real wages and capital rentals, and the stochastic processes, the economy is described by

$$B = -\frac{w_t}{C_t}, \quad (21)$$

$$\frac{1}{C_t} = \beta E_t \frac{1}{C_{t+1}} (1 + r_t^* - aB_t), \quad (22)$$

$$\frac{1}{\beta} [1 + \kappa (K_{t+1} - K_t)] = E_t \frac{C_t}{C_{t+1}} [r_{t+1} + (1 - \delta) + \kappa (K_{t+2} - K_{t+1})], \quad (23)$$

$$B_t + K_{t+1} + C_t = w_t H_t + r_t K_t + (1 - \delta) K_t - \frac{\kappa}{2} (K_{t+1} - K_t)^2 + (1 + r_{t-1}^*) B_{t-1} - aB_{t-1}^2, \quad (24)$$

$$r_t = \theta \lambda_t K_t^{\theta-1} H_t^{1-\theta}, \quad (25)$$

$$w_t = (1 - \theta) \lambda_t K_t^\theta H_t^{-\theta}, \quad (26)$$

$$\lambda_{t+1} = \gamma \lambda_t + \varepsilon_{t+1}, \quad (27)$$

$$r_t^* = \varphi r^* + (1 - \varphi) r_{t-1}^* + \varepsilon_t^r. \quad (28)$$

1.5 The Stationary State

We first determine the steady-state values of the exogenous variables. For technology,

$$\bar{\lambda} = \gamma \bar{\lambda} + E[\varepsilon_{t+1}], \quad (29)$$

and since $E[\varepsilon_{t+1}] = 1 - \gamma$,

$$\bar{\lambda} = \gamma \bar{\lambda} + (1 - \gamma) \Rightarrow \bar{\lambda} = 1. \quad (30)$$

For the foreign interest rate,

$$\bar{r}^* = \varphi r^* + (1 - \varphi) \bar{r}^* + E[\varepsilon_t^r], \quad (31)$$

and assuming $E[\varepsilon_t^r] = 0$,

$$\bar{r}^* = \varphi r^* + (1 - \varphi) \bar{r}^* \Rightarrow \bar{r}^* = r^*. \quad (32)$$

Using these values and imposing $X_s = \bar{X}$ for every s and for every $X \in \{C, K, H, B, r, w\}$, the stationary-state system becomes

$$B = -\frac{\bar{w}}{\bar{C}}, \quad (33)$$

$$\frac{1}{\beta} = 1 + r^* - a\bar{B}, \quad (34)$$

$$\frac{1}{\beta} = \bar{r} + 1 - \delta, \quad (35)$$

$$\bar{C} = \bar{w}\bar{H} + (\bar{r} - \delta)\bar{K} + r^*\bar{B} - a\bar{B}^2, \quad (36)$$

$$\bar{r} = \theta\bar{K}^{\theta-1}\bar{H}^{1-\theta}, \quad (37)$$

$$\bar{w} = (1 - \theta)\bar{K}^\theta\bar{H}^{-\theta}. \quad (38)$$

2 The Log-Linear Version

2.1 Uhlig's Method

We log-linearize the model around the stationary state using Uhlig's method. For any positive variable X_t we write

$$X_t = \bar{X} e^{\tilde{X}_t}, \quad \tilde{X}_t \approx \frac{X_t - \bar{X}}{\bar{X}}, \quad (39)$$

so that $\tilde{X}_t$ is the log-deviation of X_t from its steady-state value. To first order we use $e^{\tilde{X}_t} \approx 1 + \tilde{X}_t$ and drop any product of two deviations, since these are second order. Two rules follow and are applied throughout:

- A product of powers linearizes into a weighted sum of deviations, with the exponents as weights:

$$A_t^\alpha B_t^\beta \longrightarrow \alpha \tilde{A}_t + \beta \tilde{B}_t. \quad (40)$$

- A sum linearizes into a combination of deviations weighted by the steady-state levels:

$$A_t + B_t \longrightarrow \bar{A}\tilde{A}_t + \bar{B}\tilde{B}_t. \quad (41)$$

In each case the steady-state relation cancels the constant terms, leaving an expression that is linear in the deviations.

2.2 Intratemporal Conditions

Three equations of the model are static relations among period- t variables and linearize immediately with the product-of-powers rule. The labor-supply condition (21), $B = -w_t/C_t$, is a ratio of two variables, so

$$\tilde{w}_t - \tilde{C}_t = 0. \quad (42)$$

The rental rate (25), $r_t = \theta \lambda_t K_t^{\theta-1} H_t^{1-\theta}$, linearizes into

$$\tilde{r}_t = \tilde{\lambda}_t + (\theta - 1)\tilde{K}_t + (1 - \theta)\tilde{H}_t, \quad (43)$$

and the wage (26), $w_t = (1 - \theta)\lambda_t K_t^\theta H_t^{-\theta}$, into

$$\tilde{w}_t = \tilde{\lambda}_t + \theta\tilde{K}_t - \theta\tilde{H}_t. \quad (44)$$

2.3 The Euler equations

The two Euler equations carry expectations and require more care, since they contain the sums that involve the interest rate and the adjustment-cost terms.

Bond Euler equation

Start from (22),

$$\frac{1}{C_t} = \beta E_t \frac{1}{C_{t+1}} (1 + r_t^* - aB_t). \quad (45)$$

Substituting $C_t = \bar{C}e^{\tilde{C}_t}$, $C_{t+1} = \bar{C}e^{\tilde{C}_{t+1}}$, $r_t^* = \bar{r}^*e^{\tilde{r}_t^*}$ and $B_t = \bar{B}e^{\tilde{B}_t}$, and multiplying through by $\bar{C}$,

$$e^{-\tilde{C}_t} = \beta E_t \left[\left(1 + \bar{r}^*e^{\tilde{r}_t^*} - a\bar{B}e^{\tilde{B}_t} \right) e^{-\tilde{C}_{t+1}} \right]. \quad (46)$$

Using $e^{-\tilde{C}_t} \approx 1 - \tilde{C}_t$, $e^{\tilde{r}_t^*} \approx 1 + \tilde{r}_t^*$, $e^{\tilde{B}_t} \approx 1 + \tilde{B}_t$ and $e^{-\tilde{C}_{t+1}} \approx 1 - \tilde{C}_{t+1}$, and dropping products of deviations,

$$1 - \tilde{C}_t = \beta E_t \left[1 + \bar{r}^* - a\bar{B} + \bar{r}^*\tilde{r}_t^* - a\bar{B}\tilde{B}_t - (1 + \bar{r}^* - a\bar{B})\tilde{C}_{t+1} \right]. \quad (47)$$

The steady-state relation (34), $1 + \bar{r}^* - a\bar{B} = 1/\beta$ (with $\bar{r}^* = r^*$), cancels the constant terms, and the coefficient on $\tilde{C}_{t+1}$ becomes $\beta \cdot \frac{1}{\beta} = 1$. This leaves

$$\tilde{C}_t - E_t \tilde{C}_{t+1} + \beta \bar{r}^* \tilde{r}_t^* - \beta a \bar{B} \tilde{B}_t = 0. \quad (48)$$

Capital Euler equation

Start from (23),

$$\frac{1}{\beta} [1 + \kappa (K_{t+1} - K_t)] = E_t \frac{C_t}{C_{t+1}} [r_{t+1} + (1 - \delta) + \kappa (K_{t+2} - K_{t+1})]. \quad (49)$$

Substitute $K_t = \bar{K}e^{\tilde{K}_t}$ (and likewise for K_{t+1}, K_{t+2}), $C_t = \bar{C}e^{\tilde{C}_t}$, $C_{t+1} = \bar{C}e^{\tilde{C}_{t+1}}$ and $r_{t+1} = \bar{r}e^{\tilde{r}_{t+1}}$. Using the first-order approximations and $K_{t+1} - K_t \approx \bar{K}(\tilde{K}_{t+1} - \tilde{K}_t)$ for the adjustment-cost differences,

$$\frac{1}{\beta} [1 + \kappa \bar{K} (\tilde{K}_{t+1} - \tilde{K}_t)] = E_t \left[(1 + \tilde{C}_t - \tilde{C}_{t+1}) (\bar{r} + \bar{r}\tilde{r}_{t+1} + (1 - \delta) + \kappa \bar{K} (\tilde{K}_{t+2} - \tilde{K}_{t+1})) \right]. \quad (50)$$

Expanding the right-hand side and dropping products of deviations,

$$\frac{1}{\beta} + \frac{\kappa \bar{K}}{\beta} (\tilde{K}_{t+1} - \tilde{K}_t) = E_t \left[\bar{r} + (1 - \delta) + \bar{r}\tilde{r}_{t+1} + \kappa \bar{K} (\tilde{K}_{t+2} - \tilde{K}_{t+1}) + (\bar{r} + 1 - \delta) (\tilde{C}_t - \tilde{C}_{t+1}) \right]. \quad (51)$$

The steady-state relation (35), $\bar{r} + 1 - \delta = 1/\beta$, cancels the constant terms, and the coefficient on $\tilde{C}_t - \tilde{C}_{t+1}$ becomes $1/\beta$. Multiplying through by β and collecting terms,

$$\beta \bar{r} E_t \tilde{r}_{t+1} + \tilde{C}_t - E_t \tilde{C}_{t+1} + \beta \kappa \bar{K} E_t \tilde{K}_{t+2} - (1 + \beta) \kappa \bar{K} \tilde{K}_{t+1} + \kappa \bar{K} \tilde{K}_t = 0. \quad (52)$$

2.4 The aggregate budget constraint

Start from (24),

$$B_t + K_{t+1} + C_t = w_t H_t + r_t K_t + (1 - \delta) K_t - \frac{\kappa}{2} (K_{t+1} - K_t)^2 + (1 + r_{t-1}^*) B_{t-1} - a B_{t-1}^2. \quad (53)$$

Substitute each variable by its steady-state value times its exponential deviation. For the products $w_t H_t$ and $r_t K_t$ we use $e^{\tilde{w}_t + \tilde{H}_t} \approx 1 + \tilde{w}_t + \tilde{H}_t$ and $e^{\tilde{r}_t + \tilde{K}_t} \approx 1 + \tilde{r}_t + \tilde{K}_t$; for the quadratic debt term $B_{t-1}^2 = \bar{B}^2 e^{2\tilde{B}_{t-1}}$ we use $e^{2\tilde{B}_{t-1}} \approx 1 + 2\tilde{B}_{t-1}$. This gives

$$\begin{aligned} \bar{B} (1 + \tilde{B}_t) + \bar{K} (1 + \tilde{K}_{t+1}) + \bar{C} (1 + \tilde{C}_t) &= \bar{w} \bar{H} (1 + \tilde{w}_t + \tilde{H}_t) \\ &\quad + \bar{r} \bar{K} (1 + \tilde{r}_t + \tilde{K}_t) + (1 - \delta) \bar{K} (1 + \tilde{K}_t) \\ &\quad + \bar{B} [1 + \bar{r}^* + (1 + \bar{r}^*) \tilde{B}_{t-1} + \bar{r}^* \tilde{r}_{t-1}^*] - a \bar{B}^2 (1 + 2\tilde{B}_{t-1}). \end{aligned} \quad (54)$$

The constant terms cancel by the steady-state budget constraint (36). The adjustment-cost term does not appear at first order, since $(K_{t+1} - K_t)^2 \approx \bar{K}^2 (\tilde{K}_{t+1} - \tilde{K}_t)^2$ is second order. Collecting the

deviation terms,

$$0 = \bar{B}\tilde{B}_t + \bar{K}\tilde{K}_{t+1} + \bar{C}\tilde{C}_t - \bar{w}\bar{H}(\tilde{w}_t + \tilde{H}_t) - \bar{r}\bar{K}\tilde{r}_t - (\bar{r} + 1 - \delta)\bar{K}\tilde{K}_t - \left[(1 + \bar{r}^*)\bar{B} - 2a\bar{B}^2\right]\tilde{B}_{t-1} - \bar{r}^*\bar{B}\tilde{r}_{t-1}^*. \quad (55)$$

Finally, using $\bar{r} + 1 - \delta = 1/\beta$ from (35), the coefficient on $\tilde{K}_t$ becomes $\bar{K}/\beta$, and the log-linearized budget constraint is

$$0 = \bar{B}\tilde{B}_t + \bar{K}\tilde{K}_{t+1} + \bar{C}\tilde{C}_t - \bar{w}\bar{H}(\tilde{w}_t + \tilde{H}_t) - \bar{r}\bar{K}\tilde{r}_t - \frac{\bar{K}}{\beta}\tilde{K}_t - \left[(1 + \bar{r}^*)\bar{B} - 2a\bar{B}^2\right]\tilde{B}_{t-1} - \bar{r}^*\bar{B}\tilde{r}_{t-1}^*. \quad (56)$$

2.5 The exogenous processes

Technology. Start from (27), $\lambda_{t+1} = \gamma\lambda_t + \varepsilon_{t+1}$. Substituting $\lambda_t = \bar{\lambda}e^{\tilde{\lambda}_t} \approx \bar{\lambda}(1 + \tilde{\lambda}_t)$ and likewise for λ_{t+1} ,

$$\bar{\lambda}(1 + \tilde{\lambda}_{t+1}) = \gamma\bar{\lambda}(1 + \tilde{\lambda}_t) + \varepsilon_{t+1}. \quad (57)$$

The steady-state relation $\bar{\lambda} = \gamma\bar{\lambda} + E[\varepsilon_{t+1}]$ cancels the constant terms, leaving

$$\tilde{\lambda}_{t+1} = \gamma\tilde{\lambda}_t + \frac{\varepsilon_{t+1} - E[\varepsilon_{t+1}]}{\bar{\lambda}}. \quad (58)$$

Since $\bar{\lambda} = 1$ and $E[\varepsilon_{t+1}] = 1 - \gamma$, defining the mean-zero innovation $u_{t+1}^\lambda \equiv \varepsilon_{t+1} - (1 - \gamma)$,

$$\tilde{\lambda}_{t+1} = \gamma\tilde{\lambda}_t + u_{t+1}^\lambda, \quad E_t[u_{t+1}^\lambda] = 0. \quad (59)$$

Foreign interest rate. Start from (28), $r_t^* = \varphi r^* + (1 - \varphi)r_{t-1}^* + \varepsilon_t^r$. Substituting $r_t^* = \bar{r}^*e^{\tilde{r}_t^*} \approx \bar{r}^*(1 + \tilde{r}_t^*)$ and likewise for r_{t-1}^* ,

$$\bar{r}^*(1 + \tilde{r}_t^*) = \varphi r^* + (1 - \varphi)\bar{r}^*(1 + \tilde{r}_{t-1}^*) + \varepsilon_t^r. \quad (60)$$

Using $\bar{r}^* = r^*$, the constant terms cancel and, dividing by $\bar{r}^*$, defining the mean-zero innovation $u_t^r \equiv \varepsilon_t^r/\bar{r}^*$,

$$\tilde{r}_t^* = (1 - \varphi)\tilde{r}_{t-1}^* + u_t^r, \quad E_t[u_t^r] = 0. \quad (61)$$

2.6 The log-linearized system

Collecting (42)–(61), the log-linearized model is

$$0 = \tilde{w}_t - \tilde{C}_t, \quad (62)$$

$$0 = \tilde{r}_t - \tilde{\lambda}_t + (1 - \theta)\tilde{K}_t - (1 - \theta)\tilde{H}_t, \quad (63)$$

$$0 = \tilde{w}_t - \tilde{\lambda}_t - \theta\tilde{K}_t + \theta\tilde{H}_t, \quad (64)$$

$$0 = \tilde{C}_t - E_t\tilde{C}_{t+1} + \beta\bar{r}^*\tilde{r}_t^* - \beta a\bar{B}\tilde{B}_t, \quad (65)$$

$$0 = \beta\bar{r} E_t\tilde{r}_{t+1} + \tilde{C}_t - E_t\tilde{C}_{t+1} + \beta\kappa\bar{K} E_t\tilde{K}_{t+2} - (1 + \beta)\kappa\bar{K}\tilde{K}_{t+1} + \kappa\bar{K}\tilde{K}_t, \quad (66)$$

$$0 = \bar{B}\tilde{B}_t + \bar{K}\tilde{K}_{t+1} + \bar{C}\tilde{C}_t - \bar{w}\bar{H}(\tilde{w}_t + \tilde{H}_t) - \bar{r}\bar{K}\tilde{r}_t - \frac{\bar{K}}{\beta}\tilde{K}_t \\ - \left[(1 + \bar{r}^*)\bar{B} - 2a\bar{B}^2 \right] \tilde{B}_{t-1} - \bar{r}^*\bar{B}\tilde{r}_{t-1}^*, \quad (67)$$

$$0 = \tilde{\lambda}_{t+1} - \gamma\tilde{\lambda}_t - u_{t+1}^\lambda, \quad (68)$$

$$0 = \tilde{r}_t^* - (1 - \varphi)\tilde{r}_{t-1}^* - u_t^r. \quad (69)$$

3 The Matrices for Solving the Model

3.1 State, Jump and Exogenous Vectors

Following Uhlig's method, we sort the endogenous variables into states and jumps. The state variables, inherited from the past, are collected in

$$x_t = \begin{bmatrix} \tilde{K}_{t+1} \\ \tilde{B}_t \end{bmatrix}, \quad x_{t-1} = \begin{bmatrix} \tilde{K}_t \\ \tilde{B}_{t-1} \end{bmatrix}, \quad x_{t+1} = \begin{bmatrix} \tilde{K}_{t+2} \\ \tilde{B}_{t+1} \end{bmatrix}. \quad (70)$$

This timing convention lets the capital Euler equation, which involves $\tilde{K}_{t+2}$, $\tilde{K}_{t+1}$ and $\tilde{K}_t$, be written directly in terms of x_{t+1} , x_t and x_{t-1} . The jump variables, determined within the period, are

$$y_t = \begin{bmatrix} \tilde{C}_t \\ \tilde{H}_t \\ \tilde{r}_t \\ \tilde{w}_t \end{bmatrix}. \quad (71)$$

Because the aggregate budget constraint (56) contains the lagged foreign-rate deviation $\tilde{r}_{t-1}^*$, we enlarge the vector of stochastic variables to carry that lag as a state of the shock block,

$$z_t = \begin{bmatrix} \tilde{\lambda}_t \\ \tilde{r}_t^* \\ \tilde{r}_{t-1}^* \end{bmatrix}. \quad (72)$$

3.2 The Canonical Form

The log-linearized model is written in Uhlig's canonical form,

$$0 = Ax_t + Bx_{t-1} + Cy_t + Dz_t, \quad (73)$$

$$0 = E_t [Fx_{t+1} + Gx_t + Hx_{t-1} + Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t], \quad (74)$$

$$z_{t+1} = Nz_t + u_{t+1}. \quad (75)$$

The first block (73) collects the equations without expectations: the labor-supply condition, the two factor-price conditions, and the aggregate budget constraint. The second block (74) collects the two equations that carry expectations: the bond and capital Euler equations. This gives four equations in the first block and two in the second.

3.3 First-Block Matrices

We read the coefficients of each first-block equation against the vectors

$$x_t = [\tilde{K}_{t+1}, \tilde{B}_t]', x_{t-1} = [\tilde{K}_t, \tilde{B}_{t-1}]', y_t = [\tilde{C}_t, \tilde{H}_t, \tilde{r}_t, \tilde{w}_t]' \text{ and } z_t = [\tilde{\lambda}_t, \tilde{r}_t^*, \tilde{r}_{t-1}^*]'$$

The matrices multiplying the state vectors are

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \bar{K} & \bar{B} \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 1 - \theta & 0 \\ -\theta & 0 \\ -\frac{\bar{K}}{\beta} & -[(1 + \bar{r}^*)\bar{B} - 2a\bar{B}^2] \end{bmatrix}. \quad (76)$$

The matrix multiplying the jump vector is

$$C = \begin{bmatrix} -1 & 0 & 0 & 1 \\ 0 & -(1 - \theta) & 1 & 0 \\ 0 & \theta & 0 & 1 \\ \bar{C} & -\bar{w}\bar{H} & -\bar{r}\bar{K} & -\bar{w}\bar{H} \end{bmatrix}, \quad (77)$$

and the matrix multiplying the exogenous vector is

$$D = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -\bar{r}^*\bar{B} \end{bmatrix}. \quad (78)$$

3.4 Second-Block Matrices

The second block collects the bond Euler equation (48) (first row) and the capital Euler equation (52) (second row). Reading their coefficients against x_{t+1}, x_t, x_{t-1} gives

$$F = \begin{bmatrix} 0 & 0 \\ \beta\kappa\bar{K} & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & -\beta a\bar{B} \\ -(1+\beta)\kappa\bar{K} & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 0 & 0 \\ \kappa\bar{K} & 0 \end{bmatrix}. \quad (79)$$

Reading the coefficients against the jump vectors y_{t+1}, y_t gives

$$J = \begin{bmatrix} -1 & 0 & 0 & 0 \\ -1 & 0 & \beta\bar{r} & 0 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \quad (80)$$

and against the exogenous vectors z_{t+1}, z_t ,

$$L = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad M = \begin{bmatrix} 0 & \beta\bar{r}^* & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (81)$$

3.5 Law of Motion of the Shocks

The exogenous block $z_{t+1} = Nz_t + u_{t+1}$ stacks the technology process (59), the foreign-rate process (61), and the identity that carries the lag ($\tilde{r}_t^*$ today becomes $\tilde{r}_{t-1}^*$ tomorrow). With $z_t = [\tilde{\lambda}_t, \tilde{r}_t^*, \tilde{r}_{t-1}^*]'$ and $u_{t+1} = [u_{t+1}^\lambda, u_{t+1}^r, 0]'$,

$$N = \begin{bmatrix} \gamma & 0 & 0 \\ 0 & 1 - \varphi & 0 \\ 0 & 1 & 0 \end{bmatrix}. \quad (82)$$

4 Numerical Solution and Impulse Responses

4.1 Steady-State Calibration

We solve the log-linearized system using the `llinsolve.m` routine (Uhlig's method) for two calibrations of the economy that differ only in the steady-state foreign interest rate $\bar{r}^*$, holding all remaining structural parameters fixed, following the small open economy setup developed in McCandless's textbook treatment of open economy models.¹ The first calibration, $\bar{r}^* = 0.03$, corresponds to a net lender economy ($\bar{B} > 0$); the second, $\bar{r}^* = 0.005$, corresponds to a net borrower economy ($\bar{B} < 0$). Table 1 reports the resulting steady states.

¹McCandless, G. T. (2008). *The ABCs of RBCs: An Introduction to Dynamic Macroeconomic Models*. Harvard University Press, Chapter 13.

Table 1: Steady-state values for the two calibrations considered.

	$\bar{r}^* = 0.03$	$\bar{r}^* = 0.005$
$\bar{B}$	1.9899	-0.5101
$\bar{r}$	0.0351	0.0351
$\bar{K}/\bar{H}$	37.9893	37.9893
$\bar{w}$	2.3706	2.3706
$\bar{C}$	0.9187	0.9187
$\bar{H}$	0.3262	0.3354
$\bar{K}$	12.3934	12.7417
$\bar{Y}$	1.2084	1.2424

As $\bar{r}^*$ falls, $\bar{B}$ decreases and switches sign: a lower return on foreign assets makes external borrowing relatively cheaper, so the economy holds a lower (more negative) net foreign asset position in steady state. This decline in $\bar{B}$ is accompanied by higher $\bar{K}$, $\bar{H}$, and $\bar{Y}$, consistent with households reallocating wealth away from foreign bonds and into physical capital.

4.2 Solution Matrices

The reduced-form solution matrices P , Q , R , S returned by `llinsolve.m` are, for $\bar{r}^* = 0.03$:

$$P = \begin{bmatrix} 0.9361 & 0.0140 \\ 0.2354 & 0.7665 \end{bmatrix}, \quad Q = \begin{bmatrix} 0.1325 & -0.0264 & 0.0004 \\ -0.1766 & 0.3047 & 0.0232 \end{bmatrix}, \quad (83)$$

$$R = \begin{bmatrix} 0.3986 & 0.0886 \\ -0.1073 & -0.2460 \\ -0.7087 & -0.1575 \\ 0.3986 & 0.0886 \end{bmatrix}, \quad S = \begin{bmatrix} 0.6737 & -0.0910 & 0.0027 \\ 0.9063 & 0.2528 & -0.0075 \\ 1.5800 & 0.1618 & -0.0048 \\ 0.6737 & -0.0910 & 0.0027 \end{bmatrix}, \quad (84)$$

For $\bar{r}^* = 0.005$:

$$P = \begin{bmatrix} 0.9350 & -0.0036 \\ -0.9129 & 0.7776 \end{bmatrix}, \quad Q = \begin{bmatrix} 0.1280 & -0.0043 & -0.0000 \\ 0.6285 & -0.2058 & 0.0038 \end{bmatrix}, \quad (85)$$

$$R = \begin{bmatrix} 0.4114 & -0.0242 \\ -0.1427 & 0.0674 \\ -0.7313 & 0.0431 \\ 0.4114 & -0.0242 \end{bmatrix}, \quad S = \begin{bmatrix} 0.6846 & -0.0161 & -0.0001 \\ 0.8760 & 0.0447 & 0.0003 \\ 1.5607 & 0.0286 & 0.0002 \\ 0.6846 & -0.0161 & -0.0001 \end{bmatrix}, \quad (86)$$

where the state vector is $x_t = (\tilde{K}_{t+1}, \tilde{B}_t)'$ and the shock vector is ordered $(\tilde{\lambda}_t, \tilde{r}_t^*, \tilde{r}_{t-1}^*)'$, the third element being an auxiliary construction equal to the one-period lag of $\tilde{r}_t^*$.

4.3 Impulse Responses

Figure 1: Impulse responses to a foreign interest rate shock under both calibrations.

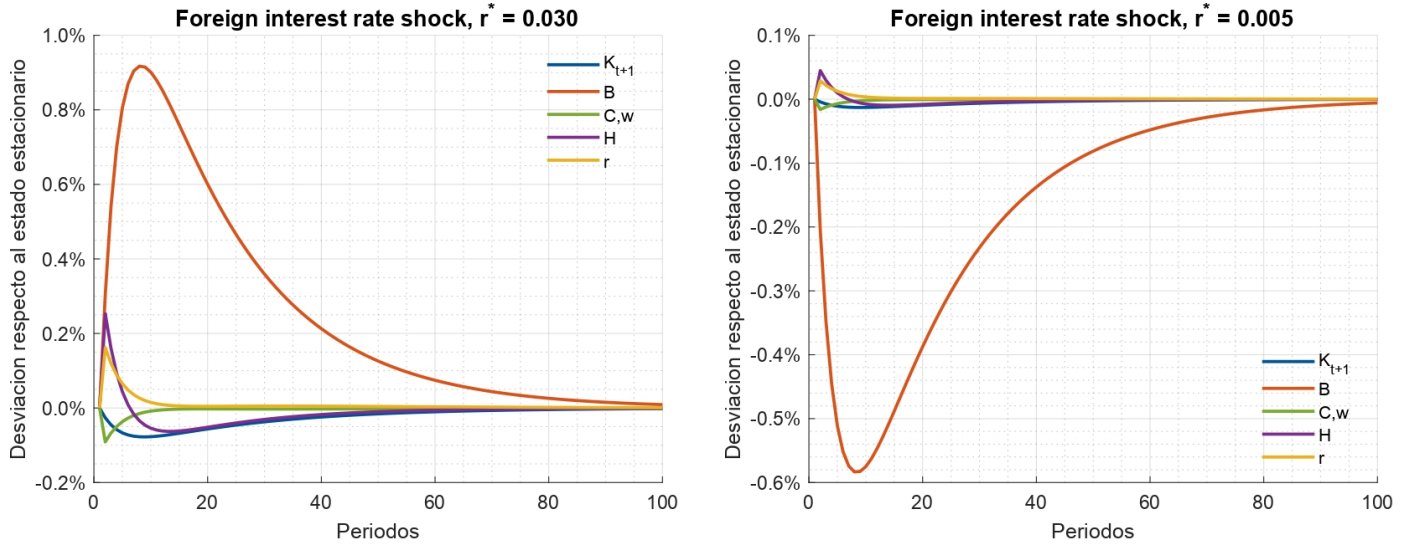


Figure 1 displays the impulse responses of $\tilde{K}_{t+1}$, $\tilde{B}_t$, $\tilde{C}_t (= \tilde{w}_t)$, $\tilde{H}_t$, and $\tilde{r}_t$ to a one-percent innovation in the foreign interest rate process, ε_t^r , for each calibration. The two calibrations generate impulse responses that differ not only in magnitude but, for $\tilde{B}_t$, in sign. This follows directly from the coefficient linking the interest-rate shock to bond accumulation in the linearized system, which is (to a first approximation) proportional to $\bar{B}$. Because $\bar{B} > 0$ when $\bar{r}^* = 0.03$ and $\bar{B} < 0$ when $\bar{r}^* = 0.005$, the same increase in r_t^* raises desired bond holdings in the first calibration but deepens net foreign debt in the second. All other variables inherit this qualitative pattern through the market-clearing and no-arbitrage conditions of the model.

4.4 MATLAB Code

The numerical solution of the model relies on three auxiliary MATLAB routines provided as part of the course material. `llinsolve.m` implements Uhlig's log-linear solution method, computing the reduced-form matrices P , Q , R , and S from the structural matrices A through N via generalized eigenvalue decomposition. `impres.m` takes these matrices together with the shock persistence matrix N and generates the impulse response functions to a shock of a specified size in a chosen innovation. `mpsolve.m` was used as a template illustrating how to call `llinsolve.m` and `impres.m` in sequence for a full model, and served as the basis for adapting the code to the open-economy setup solved here.

```
1 clear
2 clc
```

```

3 close all
4
5 beta = 0.99;
6 delta = 0.025;
7 theta = 0.36;
8 Bpar = -2.5805;
9 a = 0.01;
10 gamma = 0.95;
11 kappa = 0.20;
12 varphi = 0.05;
13
14 rstar_vec = [0.03 0.005];
15
16 for scenario = 1:length(rstar_vec)
17
18     rstar = rstar_vec(scenario);
19
20     fprintf('\n===== \n')
21     fprintf('RESULTS FOR r* = %.3f \n', rstar)
22     fprintf('===== \n')
23
24     lambdabar = 1;
25     rstarbar = rstar;
26
27     Bbar = (rstar + 1 - 1/beta) / a;
28
29     rbar = 1/beta - 1 + delta;
30
31     KHbar = (theta/rbar)^(1/(1-theta));
32
33     wbar = (1-theta) * (theta/rbar)^(theta/(1-theta));
34
35     Cbar = -wbar/Bpar;
36
37     Hbar = (Cbar - rstar*Bbar + a*Bbar^2) / ...
38           (wbar + (rbar-delta)*KHbar);
39
40     Kbar = KHbar * Hbar;
41
42     Ybar = lambdabar * Kbar^theta * Hbar^(1-theta);
43
44     fprintf('\nSteady state: \n')
45     fprintf('Bbar = %.6f \n', Bbar)
46     fprintf('rbar = %.6f \n', rbar)
47     fprintf('KHbar = %.6f \n', KHbar)
48     fprintf('wbar = %.6f \n', wbar)
49     fprintf('Cbar = %.6f \n', Cbar)

```

```

50 fprintf('Hbar = %.6f\n', Hbar)
51 fprintf('Kbar = %.6f\n', Kbar)
52 fprintf('lambdabar = %.6f\n', lambdabar)
53 fprintf('rstarbar = %.6f\n', rstarbar)
54 fprintf('Ybar = %.6f\n', Ybar)
55
56 A = [0      0;
57      0      0;
58      0      0;
59      Kbar   Bbar];
60
61 B = [0      0;
62      1-theta 0;
63      -theta  0;
64      -Kbar/beta -((1+rstarbar)*Bbar - 2*a*Bbar^2)];
65
66 C = [-1  0      0  1;
67      0  -(1-theta) 1  0;
68      0  theta      0  1;
69      Cbar -wbar*Hbar -rbar*Kbar -wbar*Hbar];
70
71 D = [0  0  0;
72      -1 0  0;
73      -1 0  0;
74      0  0 -rstarbar*Bbar];
75
76 F = [0      0;
77      beta*kappa*Kbar 0];
78
79 G = [0      -beta*a*Bbar;
80      -(1+beta)*kappa*Kbar 0];
81
82 H = [0      0;
83      kappa*Kbar 0];
84
85 J = [-1  0  0      0;
86      -1  0  beta*rbar 0];
87
88 K = [1  0  0  0;
89      1  0  0  0];
90
91 L = zeros(2,3);
92
93 M = [0  beta*rstarbar 0;
94      0  0             0];
95
96 N = [gamma  0      0;

```



```

97         0         1-varphi    0;
98         0         1           0];
99
100 [P,Q,R,S] = llinsolve(A,B,C,D,F,G,H,J,K,L,M,N,0);
101
102 fprintf('\nP =\n')
103 disp(P)
104
105 fprintf('Q =\n')
106 disp(Q)
107
108 fprintf('R =\n')
109 disp(R)
110
111 fprintf('S =\n')
112 disp(S)
113
114 figure(scenario)
115 set(gcf,'ToolBar','none')
116 set(gcf,'MenuBar','none')
117 z_r = impres(P,Q,R,S,N,1:5,2,100);
118
119 ax = gca;
120 lines = flipud(findobj(ax,'Type','Line'));
121 palette = [0.00 0.32 0.60;
122            0.85 0.33 0.10;
123            0.47 0.67 0.19;
124            0.49 0.18 0.56;
125            0.93 0.69 0.13];
126 for k = 1:length(lines)
127     lines(k).YData = lines(k).YData * 100;
128     set(lines(k),'Color',palette(k,:), 'LineWidth',1.8)
129 end
130
131 set(ax,'Color','white','FontName','Helvetica','FontSize',11,'Box','off')
132 grid(ax,'on'); grid(ax,'minor')
133 ytickformat(ax,'% .1f%%')
134 xlabel('Periodos')
135 ylabel('Desviacion respecto al estado estacionario')
136 legend('K_{t+1}','B','C,w','H','r','Location','best','Box','off')
137
138 title(sprintf('Foreign interest rate shock, r^* = %.3f', rstar), ...
139        'FontWeight','bold','FontSize',13)
140
141 if scenario == 1
142     exportgraphics(gcf,'irf_rstar_r0.03.pdf','ContentType','vector')
143 else

```

```

144     exportgraphics(gcf,'irf_rstar_r0.005.pdf','ContentType','vector')
145 end
146
147 if scenario == 2
148
149     figure
150     set(gcf,'ToolBar','none')
151     set(gcf,'MenuBar','none')
152     z_endog = impres(P,Q,R,S,N,1:5,2,100);
153
154     ax = gca;
155     lines = flipud(findobj(ax,'Type','Line'));
156     palette = [0.00 0.32 0.60;
157               0.85 0.33 0.10;
158               0.47 0.67 0.19;
159               0.49 0.18 0.56;
160               0.93 0.69 0.13];
161     for k = 1:length(lines)
162         lines(k).YData = lines(k).YData * 100;
163         set(lines(k),'Color',palette(k,:), 'LineWidth',1.8)
164     end
165
166     set(ax,'Color','white','FontName','Helvetica','FontSize',11,'Box','off')
167     grid(ax,'on'); grid(ax,'minor')
168     ytickformat(ax,'%0.1f%%')
169     xlabel('Periodos')
170     ylabel('Desviacion respecto al estado estacionario')
171     legend('K_{t+1}','B','C,w','H','r','Location','best','Box','off')
172
173     title('Foreign interest rate shock,  $r^* = 0.005$ ', ...
174           'FontWeight','bold','FontSize',13)
175
176     exportgraphics(gcf,'irf_rstar_r0.005.pdf', ...
177                   'ContentType','vector')
178
179 end
180
181 end

```